Fusion draft: SST-38/43 Helicity and gauge toolbox

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About this fusion draft

This is an automatically assembled fusion draft: multiple SST manuscripts concatenated as chapters. It is intended for navigation and editing workflow, not as a final merged publication without manual cleanup.

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Chapter 1

SST-38 — Calculate knot helicity

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Abstract

This document presents a comprehensive framework for analyzing helicity in Swirl-String Theory (SST) knot systems. We derive key expressions for total helicity, self-linking, and mutual linking, providing a robust toolset for studying topological properties of these configurations.

Objective

Compute the total helicity $\mathcal{H}$ of a knotted/linked swirl-string configuration:

$$\boxed{\mathcal{H} = \sum_k \int_{\mathcal{C}_k} \mathbf{v}_{\circlearrowleft k} \cdot \boldsymbol{\omega}_k \, dV \ + \ \sum_{i<j} 2 \,\mathrm{Lk}_{ij}\, \Gamma_i \Gamma_j} \tag{1.1}$$

This splits naturally into:

- **Self-helicity** (per component): twist + writhe within each swirl string,
- **Mutual helicity**: pairwise contributions from topological linking between distinct strings.

1. Background (SST form)

a. Swirl velocity and vorticity

- $\mathbf{v}_{\circlearrowleft}(\mathbf{r})$: local swirl velocity,
- $\boldsymbol{\omega} = \nabla \times \mathbf{v}_{\circlearrowleft}$: vorticity.

b. Circulation (quantized)

$$\Gamma_k = \oint_{\partial S_k} \mathbf{v}_{\circlearrowleft} \cdot d\boldsymbol{\ell} \qquad (\,[\Gamma] = \mathrm{m}^2/\mathrm{s}\,) \tag{1.2}$$

In SST, Γ_k is conserved (Kelvin) and takes integer quanta on closed strings, $\Gamma_k = n_k \kappa$ (not needed for computations here, but often convenient).

c. Helicity (invariant absent reconnection)

$$\mathcal{H} = \int_V \mathbf{v}_{\circlearrowleft} \cdot \boldsymbol{\omega} \, dV \tag{1.3}$$

Helicity is a topological invariant for inviscid, incompressible flow; it changes only by reconnection or boundary flux.

2. Deriving the decomposition

Assume N disjoint, thin-core swirl strings $\mathcal{K}_1, \ldots, \mathcal{K}_N$ with tubular neighborhoods $\mathcal{C}_k$.

Step 1: Split by components

$$\mathcal{H} = \sum_{i=1}^{N} \mathcal{H}_{\mathrm{self}}^{(i)} \ + \ \sum_{i<j} \mathcal{H}_{\mathrm{mutual}}^{(i,j)}. \tag{1.4}$$

Step 2: Self-helicity of component k

$$\mathcal{H}_{\mathrm{self}}^{(k)} = \int_{\mathcal{C}_k} \mathbf{v}_{\circlearrowleft k} \cdot \boldsymbol{\omega}_k \, dV \ \approx \ \Gamma_k^2 \, \mathrm{SL}_k, \tag{1.5}$$

where $\mathrm{SL}_k = \mathrm{Tw}_k + \mathrm{Wr}_k$ (Călugăreanu–White). For a standard trefoil embedding, $\mathrm{SL}_k \approx 3$.

Step 3: Mutual helicity

$$\mathcal{H}_{\mathrm{mutual}}^{(i,j)} = 2 \,\mathrm{Lk}_{ij}\, \Gamma_i \Gamma_j, \tag{1.6}$$

with Lk_{ij} the Gauss linking number between components i and j.

Final form (algebraic and integral)

$$\boxed{\mathcal{H} = \sum_{i=1}^{N} \Gamma_i^2 \, \mathrm{SL}_i \; + \; \sum_{i<j}^{N} 2 \, \mathrm{Lk}_{ij} \, \Gamma_i \Gamma_j} \tag{1.7}$$

Equivalently,

$$\boxed{\mathcal{H} = \sum_{i=1}^{N} \int_{\mathcal{C}_i} \mathbf{v}_{\circlearrowleft i} \cdot \boldsymbol{\omega}_i \, dV \; + \; \sum_{i<j} 2 \, \mathrm{Lk}_{ij} \, \Gamma_i \Gamma_j} \tag{1.8}$$

3. Practical recipe

1. Fix the knot/link type (e.g. torus link $T(p,q)$), with $N = \gcd(p,q)$ components.
2. Estimate circulation (per component) from core data; e.g. $\Gamma \approx 2\pi r_c \, \|\mathbf{v}_{\circlearrowleft}\|$ in a solid-body core model.
3. Use SL_k and Lk_{ij} for the chosen embedding (trefoil baseline: $\mathrm{SL}_k = 3$, nearest-neighbor link $\mathrm{Lk}_{ij} = 1$).
4. Evaluate:

$$\mathcal{H} \; = \; N \, \Gamma^2 \cdot \mathrm{SL} \; + \; 2 \binom{N}{2} \Gamma^2 \, .$$

4. Example: $T(18, 27)$

- $N = \gcd(18, 27) = 9, \quad \Gamma = 2\pi r_c C_e,$
- $\mathrm{SL} = 3, \quad \binom{9}{2} = 36.$

$$\mathcal{H} = 9 \cdot \Gamma^2 \cdot 3 \; + \; 2 \cdot 36 \cdot \Gamma^2 \; = \; 27\Gamma^2 + 72\Gamma^2 \; = \; 99 \, \Gamma^2. \tag{1.9}$$

5. Canonical Stability: Helicity-Energy Correspondence

The total helicity, $\mathcal{H}$, is the topological invariant ($\mathcal{H}$-conservation). The conserved energy, $\mathcal{E}$, governs the string's dynamics ($\mathcal{E}$-conservation). Their relationship dictates the stable configurations of Swirl-String matter.

A. Energy of a Swirl-String Configuration

The total energy $\mathcal{E}$ of the induced swirl-field in the background Swirl-Space (density ρ_0) is given by the kinetic energy integral:

$$\mathcal{E} = \frac{1}{2} \int_V \rho_0 \, \mathbf{v}_{\circlearrowleft} \cdot \mathbf{v}_{\circlearrowleft} \, dV \tag{1.10}$$

For N disjoint, thin-core strings $\mathcal{K}_i$, the energy decomposes into self-energy (dominated by length and core physics) and mutual energy (dominated by interaction geometry).

i. Self-Energy $\mathcal{E}_{\mathbf{self}}^{(i)}$

The self-energy of a string $\mathcal{K}_i$ of length L_i and canonical core radius r_c requires regularization. It is logarithmically dependent on the ratio of a characteristic domain radius R_0 to r_c:

$$\mathcal{E}_{\text{self}}^{(i)} \approx \frac{\rho_0 \Gamma_i^2}{4\pi} L_i \left(\ln\left(\frac{R_0}{r_c}\right) + C_{\text{geom}} \right) \tag{1.11}$$

where C_{geom} is a constant accounting for the curvature and local field topology of the string's embedding. This term drives the string to minimize its length L_i for a fixed core radius.

ii. Mutual Energy $\mathcal{E}_{\mathbf{mutual}}^{(i,j)}$

The mutual interaction energy between two strings $\mathcal{K}_i$ and $\mathcal{K}_j$ is given by the canonical double-line integral, which is a key measure of their inductive coupling:

$$\mathcal{E}_{\text{mutual}}^{(i,j)} = \frac{\rho_0 \Gamma_i \Gamma_j}{4\pi} \oint_{\mathcal{K}_i} \oint_{\mathcal{K}_j} \frac{d\boldsymbol{\ell}_i \cdot d\boldsymbol{\ell}_j}{|\mathbf{r}_i - \mathbf{r}_j|} \tag{1.12}$$

The total energy is $\mathcal{E} = \sum_i \mathcal{E}_{\text{self}}^{(i)} + \sum_{i<j} \mathcal{E}_{\text{mutual}}^{(i,j)}$.

B. Variational Stability Principle

The stable, long-lived configurations of Swirl Strings (analogous to elementary particles or stable atomic structures) are determined by a variational principle: the equilibrium shape must minimize the total energy $\mathcal{E}$ while holding the canonical topological invariant $\mathcal{H}$ fixed.

This is formalized using the method of Lagrange multipliers:

$$\delta(\mathcal{E} - \lambda \mathcal{H}) = 0 \tag{1.13}$$

The quantity λ is the **Energy-Helicity Ratio**, serving as the Lagrange multiplier, with dimensions $[\lambda] = [\mathcal{E}/\mathcal{H}] = \text{s}^{-1}$. In the canonical SST framework, λ is directly related to a characteristic global field frequency (Ω_{global}) or velocity-squared constant, $\lambda \propto \Omega_{\text{global}} \cdot \rho_0/\Gamma_0$.

C. Canonical Interpretation of $\mathcal{H}/\mathcal{E}$

The ratio $\mathcal{H}/\mathcal{E}$ (or its inverse) provides a measure of **canonical stability** and the complexity of the topological state:

- Configurations with $\lambda \approx 0$ (low helicity for high energy) tend to be dynamically unstable or highly localized.
- Configurations with a maximal λ (maximal helicity for minimal energy) represent the most **compact and topologically robust states**—these are the prime candidates for stable fundamental particles or long-lived energy structures in the Swirl-Space. The $T(18, 27)$ link, with its large $\mathcal{H}$, represents a complex structure with significant topological charge.

Canonical Note: The minimization of $\mathcal{E}$ for fixed $\mathcal{H}$ provides the canonical derivation for the **relativistic stability** of knotted string solutions, preventing the unphysical decay of topological charge predicted by some earlier VAM formulations.

Literature Foundations

Topological helicity has evolved across multiple domains—fluid mechanics, plasma physics, quantum field theory, and most recently, Swirl-String Topology (SST). This section anchors SST in a broader theoretical context.

A. Historical Origins and Classical Helicity

The concept of helicity was first introduced by **Moffatt (1969)** as a topological measure of the linkage of vortex lines. He established its conservation in ideal fluid flows and its connection to linking numbers, forming the backbone of self and mutual helicity decomposition.

B. Knot Theory in Fluid Mechanics

Ricca (1998) and **Ricca & Berger (1996)** extended Moffatt's ideas using tools from knot theory and differential geometry. They provided models for torus knots and highlighted the importance of self-linking and Seifert surfaces in representing helicity in tangled vortex structures.

C. Reconnection-Resilient Helicity (Modern Experiments)

Recent experimental and numerical work by **Scheeler et al. (2014)** and **Salman (2017)** demonstrated that total helicity can remain conserved even under vortex reconnection events. These results directly support the invariance of $\mathcal{H}$ in SST even when local geometry changes due to string interactions.

D. Toward Quantum and String-Field Analogies

Migdal (2021) interprets vortex-sheet turbulence using a solvable string theory model, suggesting deep analogies between fluid knots and flux tubes in string field theory. In SST, where $\mathcal{H}$ is tied to field energy and topological stability, this dual interpretation offers a pathway to unifying topological charge and string embeddings.

E. Comparative Table of Helicity Forms

Domain	Helicity Definition	Remarks
Classical Fluid	$\mathcal{H} = \int \mathbf{v}_{\circlearrowleft} \cdot \boldsymbol{\omega} \, dV$	Vorticity topology; Moffatt invariant
Quantum Fluid	$\mathcal{H}_q \propto n \cdot \text{Twist} + \text{Link}$	Circulation quantized; reconnection effects
SST (Swirl-String)	$\mathcal{H} = \sum \Gamma_k^2 \mathrm{SL}_k + \sum_{i<j} 2\mathrm{Lk}_{ij} \Gamma_i \Gamma_j$	Self/mutual terms; conserved unless reconnection
String-Theoretic	$\mathcal{H} \sim \int A \wedge dA$ (Chern-Simons)	Flux-tube or D-string interpretation; field dual

Table 1.1: Comparative table of helicity forms across different domains.

Note: SST helicity behaves analogously to Chern–Simons invariants in gauge theory, where linking of field lines carries conserved topological charge.

F. Summary

Together, these references justify the structure of helicity decomposition in SST and its resilience under reconnection, while connecting it to broader topological and field-theoretic frameworks. They also motivate interpreting SST knots as fundamental dual structures in a canonical field-space.

Summary Table (SST terms)

Symbol	Meaning
$\mathbf{v}_{\circlearrowleft} \cdot \boldsymbol{\omega}$	Local helicity density
Γ	Circulation around a string core (quantized, conserved)
SL_k	Self-linking of component k (Tw + Wr)
Lk_{ij}	Gauss linking number between components i, j
$\mathcal{H}$	Total helicity (self + mutual)

Canonical notes (SST): Kelvin circulation and helicity invariance apply in an inviscid, incompressible medium; $\mathcal{H}$ changes only via reconnection or boundary flux.

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Chapter 2

SST-43 — Magnetic vector / gauge

Magnetic helicity in periodic domains: gauge conditions, existence of vector potentials, and periodic winding

Setting. Let $\Omega \subset \mathbb{R}^3$ be either (i) a bounded box with one or two pairs of opposite faces identified ("1- or 2-periodic"), or (ii) the fully periodic 3-torus $\mathbb{T}^3 = \mathbb{R}^3/\Lambda$ with lattice Λ. Let $\boldsymbol{B} : \Omega \to \mathbb{R}^3$ be a smooth magnetic field with $\nabla \cdot \boldsymbol{B} = 0$.

1. Helicity, gauge, and boundary/periodicity conditions

The (total) magnetic helicity on Ω is

$$H[\boldsymbol{A}, \boldsymbol{B}] \;=\; \int_\Omega \boldsymbol{A} \cdot \boldsymbol{B} \, \mathrm{d}V, \qquad \boldsymbol{B} = \boldsymbol{\nabla} \times \boldsymbol{A}. \tag{2.1}$$

Under a gauge transform $\boldsymbol{A} \mapsto \boldsymbol{A}' = \boldsymbol{A} + \nabla\chi$,

$$\begin{aligned} H[\boldsymbol{A}', \boldsymbol{B}] - H[\boldsymbol{A}, \boldsymbol{B}] &= \int_\Omega \nabla\chi \cdot \boldsymbol{B} \, \mathrm{d}V \;=\; \int_\Omega \nabla \cdot (\chi \boldsymbol{B}) \, \mathrm{d}V \;-\; \int_\Omega \chi \, \underbrace{\nabla \cdot \boldsymbol{B}}_{0} \, \mathrm{d}V \\ &= \int_{\partial\Omega} \chi \, \boldsymbol{B} \cdot \mathrm{d}\boldsymbol{S}. \end{aligned} \tag{2.2}$$

Hence helicity is gauge invariant if and only if the boundary flux term vanishes.

Proposition 1 (Gauge invariance in periodic/closed settings)**.** *Suppose either:*

(a) Ω is bounded with perfectly conducting boundary ($\boldsymbol{B} \cdot \boldsymbol{n}|_{\partial\Omega} = 0$), or

(b) Ω is periodic in one or two directions and the net magnetic flux *through each identified pair of faces is zero; additionally the gauge function χ respects the same periodicity,*

then H in (2.1) *is gauge invariant.*

Proof. In case (a) the surface integral in (2.2) vanishes since $\boldsymbol{B} \cdot \mathrm{d}\boldsymbol{S} = 0$ on $\partial\Omega$. In case (b), identify opposite faces with orientation; periodic χ has equal values on paired faces, while zero net flux through each pair implies the oriented surface integrals cancel pairwise. Thus the sum over $\partial\Omega$ is zero. □

Proposition 2 (Obstruction in fully periodic domains)**.** *On $\mathbb{T}^3$, a* globally periodic *vector potential $\boldsymbol{A}$ with $\boldsymbol{\nabla} \times \boldsymbol{A} = \boldsymbol{B}$ exists if and only if the mean field vanishes:*

$$\langle \boldsymbol{B} \rangle \;\equiv\; \frac{1}{|\Omega|} \int_\Omega \boldsymbol{B} \, \mathrm{d}V \;=\; \boldsymbol{0}. \tag{2.3}$$

Equivalently, any nonzero constant (harmonic) component of $\boldsymbol{B}$ produces a topological obstruction to a periodic $\boldsymbol{A}$.

Sketch. Fourier decompose on $\mathbb{T}^3$. For $k \neq 0$, $\widehat{\boldsymbol{A}}(k)$ exists with $ik \times \widehat{\boldsymbol{A}}(k) = \widehat{\boldsymbol{B}}(k)$. At $k = 0$, we would require a constant $\widehat{\boldsymbol{A}}(0)$ with $\boldsymbol{\nabla} \times \widehat{\boldsymbol{A}}(0) = \boldsymbol{0}$ producing $\widehat{\boldsymbol{B}}(0) = \boldsymbol{0}$. Thus periodic solvability demands $\widehat{\boldsymbol{B}}(0) = \langle \boldsymbol{B} \rangle = 0$. $\square$

Remark 1 (Physical dimension)**.** $[\boldsymbol{A}] = \mathrm{V\,s\,m^{-1}}$, $[\boldsymbol{B}] = \mathrm{T} = \mathrm{Wb\,m^{-2}}$. *Hence* $[\boldsymbol{A}\cdot\boldsymbol{B}] = \mathrm{Wb^2\,m^{-3}}$ *and* $[H] = \mathrm{Wb^2}$ *after integration, as standard.*

2. Periodic winding and the helicity equivalence (two-periodic case)

Consider a domain that is periodic in two directions, e.g. $\Omega = [0, L_x] \times [0, L_y] \times [0, L_z]$ with identifications in x, y (a 2-torus in the horizontal plane). Let $\widetilde{\Omega} = \mathbb{R}^2 \times [0, L_z]$ be its universal cover. Field lines of $\boldsymbol{B}$ in Ω lift to curves in $\widetilde{\Omega}$.

Definition 1 (Periodic winding of a pair of field lines)**.** *Let γ_1, γ_2 be two* lifted *field lines in $\widetilde{\Omega}$, parameterized by $z \in [0, L_z]$, and let $\boldsymbol{r}(z) = \boldsymbol{x}_1(z) - \boldsymbol{x}_2(z)$ be their horizontal separation (projected to $\mathbb{R}^2$ before modding out by periods). Define the* periodic winding increment

$$\Delta\Theta(\gamma_1, \gamma_2) \;=\; \int_0^{L_z} \frac{\boldsymbol{r}(z) \times \dot{\boldsymbol{r}}(z)}{\|\boldsymbol{r}(z)\|^2} \cdot \boldsymbol{e}_z \,\mathrm{d}z, \tag{2.4}$$

and the periodic winding *as the flux-weighted mean over pairs of field lines:*

$$\mathcal{W}_{\mathrm{per}} \;=\; \frac{1}{\Phi^2} \iint \Delta\Theta(\gamma_1, \gamma_2) \,\mathrm{d}\Phi(\gamma_1) \,\mathrm{d}\Phi(\gamma_2), \tag{2.5}$$

where Φ denotes the magnetic flux measure induced by $\boldsymbol{B}$ on a transverse cross-section.

Remark 2. *Expression* (2.4) *generalizes the pairwise angular change (linking/winding) to horizontally periodic geometry by working on the universal cover and then averaging in a flux-consistent way.*

Theorem 1 (Helicity–winding equivalence in two-periodic domains)**.** *Assume: (i) ideal evolution (frozen-in field, sufficiently smooth), (ii) two-directional periodicity with zero net flux through each periodic pair, (iii) a vector potential $\boldsymbol{A}$ in a* periodic winding gauge *compatible with the identifications. Then*

$$H \;=\; \int_\Omega \boldsymbol{A} \cdot \boldsymbol{B} \,\mathrm{d}V \;=\; \Phi^2\, \mathcal{W}_{\mathrm{per}}. \tag{2.6}$$

In particular, the right-hand side defines a gauge-invariant helicity that is conserved under ideal MHD evolution.

Proof sketch. One constructs a vector potential by solving $\boldsymbol{\nabla} \times \boldsymbol{A} = \boldsymbol{B}$ with a gauge condition that fixes the mean horizontal rotational content to coincide with the periodic pairwise winding (the "periodic winding gauge"). Using the Biot–Savart–type representation adapted to the periodic geometry on $\widetilde{\Omega}$, one shows that $\boldsymbol{A} \cdot \boldsymbol{B}$ integrates to the flux-weighted average of the pairwise angular increments, yielding (2.6). Conservation follows from ideal evolution and the boundary/periodicity assumptions (no helicity flux through identified faces). $\square$

Remark 3 (Fourier connection)**.** *On horizontally periodic domains, the periodic winding helicity admits a Fourier representation; equivalently, in spectral space the gauge choice aligns the phase relations of $\widehat{\boldsymbol{A}}(k)$ with those of $\widehat{\boldsymbol{B}}(k)$ to encode winding density, providing computational routes consistent with* (2.6).

3. Summary of conditions (existence, invariance, equivalence)

- **Gauge invariance:** Eq. (2.2) shows H is gauge invariant if (PC) $\boldsymbol{B}\cdot\boldsymbol{n} = 0$ on $\partial\Omega$, or (Per) zero net flux through each periodic face pair with periodic χ.
- **Existence of periodic $\boldsymbol{A}$:** On $\mathbb{T}^3$, a periodic $\boldsymbol{A}$ exists iff $\langle \boldsymbol{B}\rangle = \mathbf{0}$; otherwise a harmonic (mean) part obstructs periodic potentials.
- **Helicity–winding equivalence (two-periodic):** Under assumptions of Theorem 1, $H = \Phi^2 \mathcal{W}_{\text{per}}$.

4. Notes on dimensions and limiting behavior

If periodicity is removed and Ω is simply connected with $\boldsymbol{B}\cdot\boldsymbol{n} = 0$, Eq. (2.6) reduces to the classical flux-weighted total winding/ linking interpretation of helicity. The dimensionality remains $[H] = \mathrm{Wb}^2$ in all cases.

5. Rosetta Translation: Magnetic Helicity → Swirl–String Theory Canon

Mapping of variables. In Swirl–String Theory (SST), the correspondence is:

$$
\begin{aligned}
\boldsymbol{B} &\longmapsto \boldsymbol{\omega} && \text{(swirl vorticity field, } [\boldsymbol{\omega}] = \mathrm{s}^{-1}\text{)},\\
\boldsymbol{A} &\longmapsto \boldsymbol{\Psi} && \text{(swirl potential / circulation density, } [\boldsymbol{\Psi}] = \mathrm{m}^2\mathrm{s}^{-1}\text{)},\\
H = \int_\Omega \boldsymbol{A}\cdot\boldsymbol{B}\,\mathrm{d}V &\longmapsto \mathcal{H}_{\text{swirl}} = \int_\Omega \boldsymbol{\Psi}\cdot\boldsymbol{\omega}\,\mathrm{d}V, &&\\
\mathcal{W}_{\text{per}} &\longmapsto \mathcal{L}_{\circlearrowleft}^{\text{per}} && \text{(periodic swirl–linking density)},\\
\Phi = \int \boldsymbol{B}\cdot\mathrm{d}\boldsymbol{S} &\longmapsto \Gamma && \text{(circulation quantum, flux of vorticity)}.
\end{aligned}
$$

Interpretation.

- The gauge invariance conditions (Prop. 1) translate to: *Swirl helicity $\mathcal{H}_{swirl}$ is invariant under potential shifts $\boldsymbol{\Psi} \mapsto \boldsymbol{\Psi} + \nabla\chi$ whenever the net circulation across identified boundaries vanishes.*
- The obstruction in $\mathbb{T}^3$ (Prop. 2) becomes: *A global swirl potential $\boldsymbol{\Psi}$ exists iff the mean vorticity $\langle\boldsymbol{\omega}\rangle = 0$. Nonzero bias corresponds to an irreducible swirl-clock offset across the periodic foliation.*
- The periodic winding theorem (Thm. 1) translates to:

$$\mathcal{H}_{\text{swirl}} = \Gamma^2\, \mathcal{L}_{\circlearrowleft}^{\text{per}}, \tag{2.7}$$

  where $\mathcal{L}_{\circlearrowleft}^{\text{per}}$ measures the pairwise phase-winding of swirl strings in the universal cover of the periodic domain.

Dimensional check.

$$[\Gamma] = \mathrm{m}^2\,\mathrm{s}^{-1}, \quad [\mathcal{L}_{\circlearrowleft}^{\text{per}}] = 1, \quad [\mathcal{H}_{\text{swirl}}] = \mathrm{m}^4\,\mathrm{s}^{-2},$$

consistent with a conserved quadratic invariant of the swirl field.

Canonical status. Equation (2.7) is classified as:

$$\boxed{\textbf{Theorem (Rosetta)}}$$

In SST, the conserved swirl helicity $\mathcal{H}_{swirl}$ equals the square of the fundamental circulation quantum Γ multiplied by the periodic swirl-linking density $\mathcal{L}_{\circlearrowleft}^{\mathrm{per}}$.

Physical picture (analogy). Swirl helicity is the "knottedness" of swirl strings. Periodicity forces us to lift the foliation to its universal cover: then, every swirl string traces a path whose *relative winding* with others accumulates across layers. The conserved $\mathcal{H}_{\mathrm{swirl}}$ counts exactly this hidden choreography of interwoven clocks.

5.1 Numerical validation (SST units, SI)

We adopt the rigid-swirl estimate for a single core loop:

$$\Gamma \;=\; \oint \mathbf{v}\cdot d\boldsymbol{\ell} \;\approx\; 2\pi\, r_c\, C_e, \quad [\Gamma] = \mathrm{m}^2\,\mathrm{s}^{-1}.$$

With your constants $\mathbf{v}_{\circlearrowleft} = 1.09384563 \times 10^6\,\mathrm{m\,s^{-1}}$, $r_c = 1.40897017 \times 10^{-15}\,\mathrm{m}$, we obtain

$$\Gamma \approx 9.683619203 \times 10^{-9}\ \mathrm{m}^2\,\mathrm{s}^{-1}.$$

Setting a representative periodic swirl-linking density $\mathcal{L}_{\circlearrowleft}^{\mathrm{per}} = 1$ (dimensionless), the Rosetta identity

$$\mathcal{H}_{\mathrm{swirl}} \;=\; \Gamma^2\, \mathcal{L}_{\circlearrowleft}^{\mathrm{per}}$$

gives

$$\mathcal{H}_{\mathrm{swirl}} \;\approx\; 9.377248088 \times 10^{-17}\ \mathrm{m}^4\,\mathrm{s}^{-2}.$$

Dimensional check. $[\,\Gamma^2\,] = \mathrm{m}^4\,\mathrm{s}^{-2}$ and $[\,\mathcal{L}_{\circlearrowleft}^{\mathrm{per}}\,] = 1$, so $[\,\mathcal{H}_{\mathrm{swirl}}\,] = \mathrm{m}^4\,\mathrm{s}^{-2}$ as required.

Scaling notes. For N coherently linked cores with identical Γ and pairwise periodic winding density contributing additively, one expects $\mathcal{H}_{\mathrm{swirl}} \sim \Gamma^2 \sum_{i<j} \mathcal{L}_{ij}^{\mathrm{per}}$, showing quadratic growth in the circulation scale and linear growth in the effective pair count.

5.2 Sensitivity scalings

Let $r_c \mapsto \lambda r_c$ and $\mathbf{v}_{\circlearrowleft} \mapsto \mu C_e$. Then

$$\Gamma(\lambda,\mu) = 2\pi(\lambda r_c)(\mu C_e) = (\lambda\mu)\,\Gamma_0, \qquad \mathcal{H}_{\mathrm{swirl}}(\lambda,\mu) = \Gamma(\lambda,\mu)^2\, \mathcal{L}_{\circlearrowleft}^{\mathrm{per}} = (\lambda\mu)^2\, \Gamma_0^2\, \mathcal{L}_{\circlearrowleft}^{\mathrm{per}}.$$

Hence $\partial \ln \mathcal{H}_{\mathrm{swirl}} / \partial \ln \lambda = 2$ and likewise for μ; the invariant is *quadratic* in each scale.

Numerics (with $\mathcal{L}_{\circlearrowleft}^{\mathrm{per}} = 1$). Using $C_e = 1.09384563 \times 10^6\,\mathrm{m\,s^{-1}}$, $r_c = 1.40897017 \times 10^{-15}\,\mathrm{m}$, we have

$$\Gamma_0 \approx 9.683619203 \times 10^{-9}\ \mathrm{m}^2\,\mathrm{s}^{-1}, \quad \Gamma_0^2 \approx 9.377248088 \times 10^{-17}\ \mathrm{m}^4\,\mathrm{s}^{-2}.$$

A grid over $\lambda, \mu \in \{0.1, 0.5, 1, 2, 5, 10\}$ confirms $\mathcal{H}_{\mathrm{swirl}} \propto (\lambda\mu)^2$.

5.3 Multi-core aggregate (uniform pairwise winding)

For N identical cores with uniform pairwise periodic winding $L_{ij} = 1$,

$$\mathcal{H}_{\text{swirl}}^{(\text{total})} = \Gamma_0^2 \sum_{i<j} L_{ij} = \Gamma_0^2 \frac{N(N-1)}{2}.$$

This shows linear growth in the number of interacting pairs and quadratic growth in the circulation scale.

Numerics. For $N = 30$ and $L_{ij} = 1$,

$$\mathcal{H}_{\text{swirl}}^{(\text{total})} = \Gamma_0^2 \frac{30 \cdot 29}{2} \approx 4.078 \times 10^{-14}\ \text{m}^4\,\text{s}^{-2}.$$

Generalization (non-uniform topology). If L_{ij} depends on geometry (knot type, relative phase, spacing), replace 1 with the measured/estimated L_{ij}:

$$\mathcal{H}_{\text{swirl}}^{(\text{total})} = \Gamma_0^2 \sum_{i<j} L_{ij}.$$

In trefoil or Hopf-linked arrays, L_{ij} can deviate from 1 due to phase slippage across periods; the periodic-winding construction (§2) provides the correct L_{ij} on the universal cover.

6. Corollaries (SST extensions beyond original works)

[Numerical grounding of circulation quantum] With $C_e = 1.09384563 \times 10^6\,\text{m}\,\text{s}^{-1}$ and $r_c = 1.40897017 \times 10^{-15}\,\text{m}$, the fundamental circulation quantum is

$$\Gamma_0 = 2\pi r_c \mathbf{v}_{\circlearrowleft} \ \approx\ 9.68 \times 10^{-9}\ \text{m}^2\,\text{s}^{-1},$$

yielding a base swirl helicity scale $\mathcal{H}_{\text{swirl}} = \Gamma_0^2 \approx 9.38 \times 10^{-17}\ \text{m}^4\,\text{s}^{-2}$.

[Quadratic scaling in core radius and swirl speed] Under the rescalings $r_c \mapsto \lambda r_c$, $C_e \mapsto \mu C_e$,

$$\mathcal{H}_{\text{swirl}}(\lambda, \mu) = (\lambda\mu)^2\, \Gamma_0^2\, \mathcal{L}_{\circlearrowleft}^{\text{per}}.$$

Thus helicity grows quadratically with either scale, providing explicit control parameters.

[Multi-core helicity growth] For N identical cores with uniform pairwise periodic swirl-linking $L_{ij} = 1$,

$$\mathcal{H}_{\text{swirl}}^{(\text{total})} = \Gamma_0^2\, \frac{N(N-1)}{2}.$$

This shows quadratic growth in Γ and combinatorial growth in N, predicting rapid amplification in linked-knot arrays or coil bundles.

[Canonical SST theorem status] Equation

$$\mathcal{H}_{\text{swirl}} \ =\ \Gamma^2\, \mathcal{L}_{\circlearrowleft}^{\text{per}}$$

is elevated from an MHD identity to a *Theorem in SST Canon*, classifying swirl helicity as a conserved invariant of foliation dynamics in periodic domains. This provides a direct bridge from topological MHD helicity to the SST energy sector ρ_E and establishes the Rosetta dictionary (potential $\mathbf{\Psi}$, vorticity $\boldsymbol{\omega}$, circulation Γ).

[Helicity–energy constitutive law for SST experiments] There exists a constitutive coupling between conserved swirl helicity and the energy density (swirl pressure) of the foliation,

$$\rho_E \ =\ \kappa_H\, \frac{\rho_f}{L_H^2}\, \mathcal{H}_{\text{swirl}}, \qquad \mathcal{H}_{\text{swirl}} \ =\ \Gamma^2\, \mathcal{L}_{\circlearrowleft}^{\text{per}}, \tag{C5}$$

where ρ_f is the effective (fluid) density, κ_H is a dimensionless calibration constant, L_H is a helicity coherence length (the transverse scale over which pairwise periodic winding coherently contributes), and $\mathcal{L}_{\circlearrowleft}^{\text{per}}$ is the periodic swirl-linking density of Sec. 5.

Status. *Constitutive (from Canon + calibration).* The identity $\mathcal{H}_{\text{swirl}} = \Gamma^2 \mathcal{L}_{\circlearrowleft}^{\text{per}}$ is Theorem (Rosetta). The proportionality to ρ_E is a phenomenological law constrained by units and validated by experiment/simulation via κ_H and L_H.

Dimensional audit. $[\rho_f] = \mathrm{kg\,m^{-3}}$, $[\mathcal{H}_{\text{swirl}}] = \mathrm{m^4\,s^{-2}}$, $[L_H^{-2}] = \mathrm{m^{-2}}$. Thus $[\rho_E] = \mathrm{kg\,m^{-1}\,s^{-2}} = \mathrm{J\,m^{-3}}$ (a pressure), as required.

Scalings and limits. Using $\Gamma = 2\pi r_c C_e$ and (C5),

$$\rho_E = \kappa_H \frac{\rho_f}{L_H^2} (2\pi r_c C_e)^2 \mathcal{L}_{\circlearrowleft}^{\text{per}} \propto \rho_f \frac{r_c^2 C_e^2}{L_H^2} \mathcal{L}_{\circlearrowleft}^{\text{per}}.$$

Hence $\partial \ln \rho_E / \partial \ln r_c = 2$, $\partial \ln \rho_E / \partial \ln \mathbf{v}_{\circlearrowleft} = 2$, $\partial \ln \rho_E / \partial \ln L_H = -2$, and ρ_E grows linearly with $\mathcal{L}_{\circlearrowleft}^{\text{per}}$. For N identical cores with uniform $L_{ij} = 1$, $\mathcal{L}_{\circlearrowleft}^{\text{per}} \sim N(N-1)/2$ (see Sec. 5.3), giving rapid amplification in linked arrays.

Numerical illustration (with your constants). Let $\rho_f = 7.0 \times 10^{-7}\ \mathrm{kg\,m^{-3}}$, $r_c = 1.40897017 \times 10^{-15}$ m, $C_e = 1.09384563 \times 10^6\ \mathrm{m\,s^{-1}}$. Then $\Gamma_0 \approx 9.683619203 \times 10^{-9}\ \mathrm{m^2\,s^{-1}}$ and $\Gamma_0^2 \approx 9.377248088 \times 10^{-17}\ \mathrm{m^4\,s^{-2}}$. For a single-core, unit linking density ($\mathcal{L}_{\circlearrowleft}^{\text{per}} = 1$),

$$\rho_E = \kappa_H \frac{7.0 \times 10^{-7}}{L_H^2} \left(9.377248088 \times 10^{-17}\right)\ \mathrm{J\,m^{-3}}.$$

Example values:

$$\begin{aligned}
L_H = 10^{-2}\ \mathrm{m}: &\quad \rho_E \approx \kappa_H\, 6.56 \times 10^{-19}\ \mathrm{J\,m^{-3}},\\
L_H = 10^{-4}\ \mathrm{m}: &\quad \rho_E \approx \kappa_H\, 6.56 \times 10^{-15}\ \mathrm{J\,m^{-3}},\\
L_H = 10^{-6}\ \mathrm{m}: &\quad \rho_E \approx \kappa_H\, 6.56 \times 10^{-11}\ \mathrm{J\,m^{-3}}.
\end{aligned}$$

For $N = 30$ with uniform $L_{ij} = 1$ ($\mathcal{L}_{\circlearrowleft}^{\text{per}} = 435$), multiply these by 435.

From energy density to thrust. In the Euler–SST sector, swirl pressure equals energy density: $p_{\text{swirl}} = \rho_E$. A directed pressure gradient produces force $F \approx \Delta p\, A = \Delta \rho_E\, A$ on area A. Thus asymmetric control of L_H and/or $\mathcal{L}_{\circlearrowleft}^{\text{per}}$ (e.g. phase-biased linking across the device) yields a net thrust:

$$F \approx \kappa_H \frac{\rho_f}{L_H^2} \Delta(\Gamma^2 \mathcal{L}_{\circlearrowleft}^{\text{per}})\, A.$$

Calibration protocol (minimal experiment).

1. Build a multi-core array with controllable pairwise phase to tune $\mathcal{L}_{\circlearrowleft}^{\text{per}}$.
2. Fix (r_c, C_e) drive; vary N and the phase program to scan $\mathcal{L}_{\circlearrowleft}^{\text{per}}$.
3. Measure net force F on a thrust stand while modulating a boundary layer to set L_H (e.g. dielectric spacing or flow-ring spacing).
4. Fit κ_H and L_H via $F \approx (\rho_f / L_H^2)\, \kappa_H\, \Delta(\Gamma^2 \mathcal{L}_{\circlearrowleft}^{\text{per}})\, A$.

This closes the loop from the conserved topological invariant to an experimentally measurable propulsion observable.